

Norm continuity of the $\mathrm{GL}(n)$ action in low degree and in dimension two

A substantial partial result for arXiv:2401.05913

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Status: substantial partial result, likely valid. The action is norm-continuous on all valuations when $n = 2$ and on the degree-zero and degree-one summands in every dimension. For $n \geq 3$, only a possibly nonsmooth degree-two summand remains.

1 Source question and notation

Let S^{n-1} be the Euclidean unit sphere and equip $\mathrm{Lip}(S^{n-1})$ with

$$\|f\|_{\mathrm{Lip}} = \max\{\|f\|_{\infty}, \mathrm{lip}(f)\}.$$

Write $\mathcal{V}_n$ for the space of complex-valued, τ -continuous, dually translation invariant valuations on $\mathrm{Lip}(S^{n-1})$. Thus $\mu(f + \ell) = \mu(f)$ whenever ℓ is the restriction of a linear functional on $\mathbb{R}^n$. The source [1] equips this space with the Banach norm

$$\|\mu\| = \sup_{\|f\|_{\mathrm{Lip}} \leq 1} |\mu(f)|$$

and, in Remark 4.3 on PDF page 18, asks whether the natural action

$$[\pi(g)\mu](f) = \mu(g^{-1} \cdot f), \quad g \in \mathrm{GL}(n, \mathbb{R}),$$

is continuous for this norm. Here

$$(g \cdot f)(x) = |g^T x| f\left(\frac{g^T x}{|g^T x|}\right).$$

The source proves the homogeneous decomposition

$$\mathcal{V}_n = \mathcal{V}_{n,0} \oplus \mathcal{V}_{n,1} \oplus \mathcal{V}_{n,2}.$$

Our result settles the first two summands and all three when $n = 2$.

2 Idea of the proof

Restriction to support functions gives an injective equivariant map from the functional valuations to classical translation invariant valuations on convex bodies. In degree one, McMullen polynomiality says that the classical valuation is Minkowski additive, hence integration of the support function against a finite measure. The same integral defines a valuation on every Lipschitz function, and

$$\geq \frac{\varepsilon}{2} + \|\mu - \nu\|_{\tilde{K}} < \varepsilon.$$

Thus the operation is continuous. $\square$

Remark 4.3. *It was shown in [22] that every τ -continuous valuation on $\text{Lip}(S^{n-1})$ is bounded on the sets $\text{Lip}_M(S^{n-1})$, $M > 0$. In particular, one may equip $\text{Val}(\text{Lip}(S^{n-1}))$ with*

$$\|\mu\| := \sup_{f \in \text{Lip}_1(S^{n-1})} |\mu(f)|, \quad \text{for } \mu \in \text{Val}(\text{Lip}(S^{n-1})),$$

which is a norm due to the homogeneous decomposition. It is easy to see that $\text{Val}(\text{Lip}(S^{n-1}))$ is complete with respect to $\|\cdot\|$, however we do not know whether the operation of $\text{GL}(n, \mathbb{R})$ is continuous with respect to this norm. We are thus forced to work with the topology of uniform convergence on compact subsets, which is finer than the topology induced by this norm (note that every compact set is contained in $\text{Lip}_M(S^{n-1})$ for some $M > 0$, by Proposition 3.5).

4.2 Relation to translation invariant valuations on convex bodies

Recall that we have the map

$$P : \mathcal{K}(\mathbb{R}^n) \rightarrow \text{Lip}(S^{n-1})$$

Figure 1: The norm-continuity question in [1, Remark 4.3].

injectivity forces it to be the original valuation. Small linear changes move a Lipschitz function uniformly by a small amount, uniformly over the unit ball, so integration against the fixed measure gives norm-continuity.

In dimension two the degree-zero part is constant and the classical degree-two space is spanned by area. Its known functional extension $\int(f^2 - |\nabla f|^2)$ is therefore the only possible degree-two part, and it transforms by the continuous scalar $|\det g|^{-1}$. Local operator bounds then turn continuity of fixed orbits into joint continuity of the action.

3 The low-degree theorem

Let $T : \mathcal{V}_n \rightarrow \text{Val}(\mathbb{R}^n)$ be the source's injective map

$$T\mu(K) = \mu(h_K),$$

where h_K is the support function of the convex body K .

Theorem 1 (low degree and dimension two). *Let $n \geq 2$.*

1. *The $\text{GL}(n, \mathbb{R})$ action is jointly continuous in the norm of Remark 4.3 on $\mathcal{V}_{n,0} \oplus \mathcal{V}_{n,1}$.*
2. *Every $\mu \in \mathcal{V}_{n,1}$ has the form*

$$\mu(f) = \int_{S^{n-1}} f d\sigma,$$

where σ is a finite complex Borel measure satisfying $\int_{S^{n-1}} x d\sigma(x) = 0$.

3. *For $n = 2$, the action is jointly norm-continuous on all of $\mathcal{V}_2$.*

Proof. A zero-homogeneous continuous valuation is constant: for $t > 0$, $\mu_0(tf) = \mu_0(f)$, and letting $t \downarrow 0$ gives $\mu_0(f) = \mu_0(0)$. Its orbit is therefore fixed.

Now let $\mu \in \mathcal{V}_{n,1}$. The classical valuation $T\mu$ is continuous, translation invariant, and one-homogeneous. The degree-one case of McMullen polynomiality makes it Minkowski additive [2, 3].

The standard support-function representation therefore gives a finite complex Borel measure σ on S^{n-1} such that

$$T\mu(K) = \int_{S^{n-1}} h_K d\sigma.$$

Translation invariance is exactly the zero-first-moment condition on σ . Define $\mu_\sigma(f) = \int f d\sigma$. This is a linear, hence one-homogeneous, valuation; it is dually translation invariant by the moment condition and τ -continuous because τ -convergence includes uniform convergence. Moreover, $T\mu_\sigma = T\mu$. Injectivity of T from [1, Proposition 4.4] yields $\mu = \mu_\sigma$. (For complex-valued valuations, apply the real representation to the real and imaginary parts.)

We prove continuity of this orbit at the identity. Put $h = g^{-1}$ and

$$\Phi_h(x) = \frac{h^T x}{|h^T x|}.$$

Uniformly over $x \in S^{n-1}$ and $\|f\|_{\text{Lip}} \leq 1$,

$$\begin{aligned} |(h \cdot f)(x) - f(x)| &\leq |h^T x| - 1 |f(\Phi_h(x))| + |f(\Phi_h(x)) - f(x)| \\ &\leq \|h - I\|_{\text{op}} + |\Phi_h(x) - x|. \end{aligned}$$

The right side tends uniformly to zero as $g \rightarrow I$. Consequently,

$$\|\pi(g)\mu - \mu\| \leq \|\sigma\|_{\text{TV}} \sup_{\|f\|_{\text{Lip}} \leq 1} \|h \cdot f - f\|_\infty \longrightarrow 0.$$

Suppose next that $n = 2$. The map T embeds $\mathcal{V}_{2,2}$ into the one-dimensional classical space $\text{Val}_2(\mathbb{R}^2)$ spanned by area. The nonzero valuation

$$Q(f) = \int_{S^1} (f(x)^2 - |\nabla f(x)|^2) d\mathcal{H}^1(x)$$

belongs to $\mathcal{V}_{2,2}$ and restricts to a nonzero multiple of area [4, 1]. Hence $\mathcal{V}_{2,2} = \mathbb{C}Q$. Equivariance and injectivity of T give

$$\pi(g)Q = |\det g|^{-1}Q,$$

so this final orbit is norm-continuous.

For completeness, the representation operators are locally bounded. The source's gradient covariance implies

$$\|g \cdot f\|_{\text{Lip}} \leq 2\|g\|_{\text{op}}\|f\|_{\text{Lip}}.$$

The homogeneous projections are bounded as well: writing $A = \mu(f/2) - \mu(0)$ and $B = \mu(f) - \mu(0)$ gives

$$\mu_2(f) = 2B - 4A, \quad \mu_1(f) = 4A - B,$$

and therefore $\|\mu_k\| \leq 12\|\mu\|$ (a deliberately uniform bound for $k = 0, 1, 2$). Positive homogeneity now gives locally uniform bounds for $\pi(g)$. Continuity of every fixed orbit proved above consequently implies joint continuity. $\square$

4 An integral degree-two extension

The same mechanism covers a broad part of the remaining degree-two space. Write

$$\bar{\nabla}f(x) = \nabla f(x) + f(x)x.$$

Proposition 2 (integral degree-two class). *Let $\varphi \in L^1(S^{n-1}; \text{Sym}^2(\mathbb{C}^n))$ satisfy*

$$\int_{S^{n-1}} \varphi(x) \bar{\nabla} f(x) d\mathcal{H}^{n-1}(x) = 0 \quad (f \in \text{Lip}(S^{n-1})).$$

Then

$$\mu_\varphi(f) = \int_{S^{n-1}} \langle \varphi(x) \bar{\nabla} f(x), \bar{\nabla} f(x) \rangle d\mathcal{H}^{n-1}(x)$$

belongs to $\mathcal{V}_{n,2}$, and its $\text{GL}(n, \mathbb{R})$ orbit is norm-continuous. The same holds for the norm closure of all such valuations.

Proof. At almost every point, the unordered pair of first jets of $f \vee h$ and $f \wedge h$ equals that of f and h ; on the equality set their gradients agree almost everywhere. This proves the valuation identity. Homogeneity is immediate. If $\ell_a(x) = \langle a, x \rangle$, then $\bar{\nabla} \ell_a = a$. Expanding $\mu_\varphi(f + \ell_a)$ and using the stated condition, first for f and then for linear functions, proves dual translation invariance. For a τ -convergent sequence the augmented gradients converge almost everywhere and are uniformly bounded; dominated convergence against the fixed L^1 coefficient proves τ -continuity. Also

$$\|\mu_\varphi\| \leq 4\|\varphi\|_{L^1}.$$

For $h = g^{-1}$, the source's covariance formula reads

$$\bar{\nabla}(h \cdot f)(x) = h \bar{\nabla} f(\Phi_h(x)).$$

Changing variables $y = \Phi_h(x)$ rewrites $\pi(g)\mu_\varphi$ as μ_{φ_g} , where, with J_g the spherical Jacobian of Φ_g ,

$$\varphi_g(y) = J_g(y) h^T \varphi(\Phi_g(y)) h.$$

As $g \rightarrow I$, the matrix factors and J_g converge uniformly and pullback by Φ_g is strongly continuous on L^1 . Thus $\|\varphi_g - \varphi\|_1 \rightarrow 0$ and

$$\|\pi(g)\mu_\varphi - \mu_\varphi\| \leq 4\|\varphi_g - \varphi\|_1 \rightarrow 0.$$

Local boundedness of the representation passes this conclusion to the norm closure. $\square$

Every smooth degree-two valuation classified in [1, Theorem 3] has precisely the form in Proposition 2 with a smooth matrix field. The proposition therefore includes the norm closure of the source's smooth degree-two subspace.

5 Limitations and novelty

For $n \geq 3$, the full question is now confined to degree two outside the norm closure of the integral class. The source proves density of smooth valuations only for uniform convergence on compact subsets of $\text{Lip}(S^{n-1})$; its unit ball is not compact in the τ metric because it contains arbitrarily oscillatory gradients. Convolution by a group approximate identity cannot repair this directly: norm convergence of that convolution is equivalent to the orbit continuity one is trying to prove.

The most obvious negative construction also fails. A finitely additive or singular coefficient can make translations discontinuous in a dual norm, but τ -continuity excludes finitely additive gradient functionals [4]; singular quadratic coefficients would require evaluating an almost-everywhere gradient on a null set. Thus neither a full representation nor a valid discontinuous orbit has been obtained for the residual summand.

A bounded novelty search through 11 August 2026 checked the run’s exact-id and keyword indexes, exact searches for the Remark 4.3 wording and for “GL action norm continuity” with the paper’s title and terminology, and the OpenAlex record for arXiv:2401.05913 (zero indexed citations). No later answer or matching low-degree theorem was located. Novelty confidence is moderate: the measure argument is classical once the source’s injectivity is noticed, but no prior statement of this consequence was found.

Human-review recommendation. Check especially the use of the classical degree-one measure representation, the passage from injectivity of T to the formula for all Lipschitz functions, and the L^1 change-of-variables formula in Proposition 2.

References

- [1] A. Colesanti, J. Knoerr, and D. Pagnini, *The homogeneous decomposition of dually translation invariant valuations on Lipschitz functions on the sphere*, arXiv:2401.05913 (2024).
- [2] P. McMullen, *Continuous translation invariant valuations on the space of compact convex sets*, Arch. Math. 34 (1980), 377–384.
- [3] M. Ludwig and F. Mussnig, *Valuations on convex bodies and functions*, arXiv:2302.00416 (2023), Corollary 4.9.
- [4] A. Colesanti, D. Pagnini, P. Tradacete, and I. Villanueva, *Continuous valuations on the space of Lipschitz functions on the sphere*, J. Funct. Anal. 280 (2021), 108873; arXiv:2005.05419.